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Inventor(s)	Weber; Lori

Collapsible litter box

Abstract

A litter box that is portable and collapsible providing ease of traveling with the litter box. The litter box includes a bottom tray and a top enclosure. The bottom tray is a hollow rectangle with a bottom and the top enclosure has two collapsible walls and a roof that covers the bottom tray when in the collapsed position. The litter box top enclosure is erected for the use of the litter box and the top enclosure can be collapsed for travel.

Inventors: **Weber; Lori (Land O Lakes, FL)**

Applicant: **Weber; Lori** (Land O Lakes, FL)

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Primary Examiner: Hutchens; Christopher D

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to provisional application U.S. Ser. No. 63/443,826 filed Feb. 7, 2023.

FIELD OF THE DISCLOSURE
(1) This disclosure relates to an animal litter box, and more specifically to an enclosed, portable, and collapsible animal litter box.
BACKGROUND
(2) Domesticated animals, such as cats and dogs, are enjoyed as a form of companionship. A vast majority of households in the United States include a companion animal. Pets not only provide love and affection, but they are also linked to keeping a pet owner well. Pet ownership is linked to lower blood pressure, reduced stress, and overall health. So, pets make pet owners happier and healthier.
(3) Some pets are taken on travels with their owners which may entail riding in vehicles, on both long and short rides. On long rides, pets need to relieve themselves. Often a vehicle may stop at a rest stop, or other relevant location, to allow a pet to relieve themselves. There are some pets that may not use the outdoors, such as cats, which are commonly used to a litter box. For such situations, pet owners may travel with a litter box that are for use by animals that naturally like to bury their waste or animals that may be trained to use such litter boxes. However, travelling with a litter box can cause its own issues, such as the size of the litter box can use up too much space, the litter can spill with the jostling in the vehicle, and other issues. There are disposable litter boxes that can be used, but that creates an environmental issue where excess waste in the form of the litter boxes is discarded into the waste.
(4) This disclosure provides a novel article that addresses the above issue for a portable litter box.
SUMMARY

(5) This disclosure provides a novel litter box that is portable and collapsible providing ease of traveling with the litter box. The disclosure also provides a litter box which prevents the loss of litter from the box or animal waste within the litter. The disclosure is also directed toward a litter box that can be opened for an animal to use and is an enclosed box. Additionally, the litter box is also collapsible, which can be collapsed after use so minimal space is used when being stored. Further, collapsing the box also prevents litter and associated waste from escaping the litter box.
(6) According to one aspect of this disclosure, an animal litter box is provided. The litter box comprises a bottom tray and a top enclosure. The bottom tray is configured having four upstanding walls for holding litter. The top enclosure is configured with a first side wall, a second side, and a roof that is connected to the first and second side walls. The first and second side walls are also connected to the corresponding side walls on the bottom tray. The bottom tray and the top enclosure form an enclosed space for an animal to use. The litter box also includes a back wall that is hingedly connected to the bottom tray so that the back wall can be folded down over the open the litter tray during storage.
(7) These and other aspects are depicted in the accompanying figures and described below and will be further apparent based thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of this disclosure are described in detail below with reference to the drawings. Features, aspects, and advantages of this disclosure will be better understood with reference to the following description, appended claims, and accompanying drawings. The drawings provided herein are for illustrative purposes only of selected embodiments, and not all possible implementations, and are not intended to limit the scope of this disclosure.
(2) FIG. **1** is a conceptual illustration of a perspective view of a litter box, in accordance with an illustrative embodiment.
(3) FIG. **2** is a pictorial illustration of a back side of the litter box in accordance with an illustrative embodiment.
(4) FIG. **3** is a pictorial illustration of back side of the litter box showing a back wall in a lowered position, in accordance with an illustrative embodiment.
(5) FIG. **4** is a pictorial illustration of a front view of the litter box showing the litter box being manipulated into a collapsed position, in accordance with an illustrative embodiment.
(6) FIG. **5** is a pictorial illustration of the litter box in a collapsed position, in accordance with an illustrative embodiment.
(7) FIG. **6** is a pictorial illustration of an internal side view of the litter box in a collapsed position showing a rod on an inside connected to a carrying handle which is on an exterior of the litter box, in accordance with an illustrative embodiment.
(8) FIG. **7** is a pictorial illustration of the internal side view of the litter box from FIG. **6** showing the rod opening the top enclosure while the carrying handle is lowered, in accordance with an illustrative embodiment.
(9) FIG. **8** is a pictorial illustration of the internal side view of the litter box from FIG. **6** showing the top enclosure fully opened with the rod in accordance with an illustrative embodiment.
DETAILED DESCRIPTION
(10) In this detailed description, the summary above, the claims below, and the accompanying drawings, reference is made to particular features and aspects (including method steps). It should be understood that this disclosure includes all possible combinations of such features and aspects. For example, where a particular feature is disclosed in the context of a particular embodiment or implementation, or in a particular claim, that feature may also be used, to the extent possible, in

combination with and/or in the context of other particular aspects and embodiments of this disclosure.

(11) The term "comprises" and grammatical equivalents thereof are used herein to mean that other features, components, and steps are optionally present. For example, an article "comprising" (or "which comprises") components A, B, and C can consist of (i.e., contain only) components A, B, and C, or can contain not only components A, B, and C but also contain one or more other components.

(12) Where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where the context excludes that possibility), and the method can include one or more other steps that are carried out before any of the defined steps, between two of the defined steps, or after all of the defined steps (except where the context excludes that possibility).

(13) Certain terminology and derivations thereof may be used in the following description for non-limiting purposes of convenience. For example, terms such as "opposite," "upward," "downward," "left," "right," "middle," "adjacent," and "behind" refer to directions in the drawings to which reference is made unless otherwise stated. Similarly, words such as "inward" and "outward" refer to directions toward and away from, respectively, the geometric center of a device or area and designated parts thereof. References in the singular tense include the plural, and vice versa, unless otherwise noted. Terms such as "attached to," "coupled to," "affixed to," "fastened to," etc. as used herein may refer to a direct or indirect connection.

(14) This disclosure is drawn to an animal litter box which is an enclosed, collapsible, and portable box which can be used when traveling in a vehicle. A top enclosure of the litter box is provided such that it can collapse onto a top of a bottom tray during storage which minimizes the amount of space the litter box takes and prevents litter in the bottom tray from leaking or spilling out.

(15) FIGS. **1** to **8** illustrate a conceptual diagram of a litter box **100**. FIG. **1** is a conceptual diagram of the litter box **100**, in an extended or fully opened position for use, in accordance with non-limiting embodiments of this disclosure. In the illustrated non-limiting embodiment, the litter box **100** takes a geometric form of a rectangular box in the open position and a collapsed position (see FIG. **5**). In this regard, although this disclosure is presented primarily in the context of being used as a litter box to be used by an animal, it should be understood that the litter box **100** as presented may be used in context with other type of storage and carrying boxes.

(16) As best seen in FIGS. **1** and **2**, the litter box **100** comprises of a bottom tray **110**, a top enclosure **120**, and a rear panel **130**. The bottom tray **110** of the litter box **100** is configured for holding litter and includes a base **111** with a front wall **112***a*, a back wall **112***b*, a left wall **112***c*, and a right wall **112***d*, wherein the walls define a length and a width of the bottom tray **110**. The top enclosure **120**, as best seen in an extended or open position in FIG. **1**, includes a top **121**, a left side wall **122***a*, and a right side wall **122***b*. The directions "front," "back," "left," and "right" are strictly used in conjunction with the presented view of the illustration in FIG. **1** and not intended to be limiting in any way. The top enclosure **120** may be connected to the bottom tray **110**. Additionally, the rear panel **130** may be connected to the bottom tray **110**, or alternatively may be connected to the top enclosure **120**.

(17) As mentioned above, the top enclosure **120** is connected to the bottom tray **110**. Specifically, the left side wall **122***a* and the right side wall **122***b* are connected to their respective side walls **112***c*, **112***d* on the bottom enclosure **110**. Additionally, the top panel **121** has a hinged connection **123** to each of the left side wall **122***a* and the right side wall **122***b*. The left side wall **122***a* and the right side wall **122***b* are also hingedly connected **127** to the left and right wall **112***c*, **112***d* on the bottom tray **110**. Each of these hinged connections may have one straight hinge that runs an entire length of the connection, or the hinged connections may include one or more hinges interspersed along a length of the connection. Additionally, the left side wall **122***a* and the right side wall **122***b* each have a one or more hinges **125** along their lengths at a relative centerline that separates each

of the left and right side walls **122***a*, **122***b* into two relatively equal sections.
(18) As seen in FIG. **4**, the hinged connections **123**, **125**, **127** allow the top enclosure **120** to fold down on to a top of the bottom tray **110**. The tops enclosure **120** will fold at the hinged connections **123**, **125**, **127** such that the left and right walls **122***a*, **122***b* will fold inward toward each other with the top panel **121** resting on top of the folded left and tight walls **122***a*, **122***b*. FIG. **5** illustrates the litter box **100** in a collapsed position wherein the top enclosure **120** has been collapsed, or folded down, onto the bottom tray **110** such that only the top panel **121** of the top enclosure **120** is presented. As seen in FIGS. **1** and **5**, the litter box also comprises one or more clips **114** which may be used to hold the top enclosure **120** in the collapsed position over the bottom tray **110**. As seen in the illustrations, the one or more clips **114** are included on the bottom tray and may be hingedly connected such that the one or more clips **114** can be moved onto the top panel **121** when the top enclosure **120** is folded downward onto the bottom tray **110**. As shown, the one or more clips **114** are included on the right side wall **112***d* of the bottom tray **110**. It is to be understood that the one or more clips **114** may be included on the left side wall **112***c*, or on both left and right side walls **112***c*, **112***d*. It is also to be understood that other mechanisms or features that can hold the top enclosure **120** in the collapsed position are also within the disclosure of this invention.
(19) FIGS. **2** and **3** illustrate a rear view of the litter box **100**. The rear wall **130** is shown to be hingedly connected **133** to the back wall **112***b* on the bottom tray **110**. The hinged connection **133** is provided to position the rear panel **130** either extending upward in line with the back wall **112***b* or pressed inward over the top of the bottom tray **110**. In use, the rear panel **130** is extending upward to present an animal with an uncovered bottom tray **110** for use. A user may press the rear panel **130** at the hinged connection **133** (see, FIG. **3**) to allow for the movement of the top enclosure **120** downward into the collapsed position. As seen in FIG. **2**, the rear panel **130** also includes a vent **132** which may provide for additional ventilation.
(20) As shown in FIG. **1**, it will be readily appreciated that a front of the litter box **100** is open and is provided as an entryway **102** when the litter box **100** is in the fully opened position. The entry way **102** defines an entry for an animal to enter into the litter box **100**. Along with the vent **132** in the rear panel **130**, proper ventilation is also provided with a crossflow of air.
(21) The litter box **100** further includes a carrying handle **140** which is provided connected to both the left and right side walls **112***c*, **112***d*. As best seen in FIGS. **1** and **5** to **8**, the carrying handle includes a latch **142** which is provided on an inside of the litter box. The carrying handle **140** and the latch **142** are perpendicular to each other and are connected to each other such that as the handle **140** is moved about a hinge at the left and right walls **112***c*, **112***d*, the latch will move respectively while remaining perpendicular to the carrying handle **140**. The latch **142** in its configuration and attachment to the carrying handle **142** allows the top enclosure **120** to be opened or collapsed. It will be appreciated that the latch **142** is positioned on each of the left and right side walls **112***c*, **112***d* and is also positioned such that when the latch **142** is in the upright position (wherein the carrying handle is parallel to the length of the bottom tray **110**), the latch **142** is abutting against the left side wall **122***a* and right side wall **122***b* of the top enclosure **120**. The concurrent movement of the carrying handle **140** and the latch **142** will be readily understood and will be explained below for better understanding.
(22) FIGS. **6** to **8** illustrate the litter box **100** with the carrying handle **140** in the various positions with the concurrent movement of the latch **142** illustrated on the interior of the litter box **100**. As seen in FIG. **6**, the carrying handle **140** is positioned upright to allow the litter box **100** to be carried and the latch **142** is positioned parallel along each of the left and right walls **112***c*, **112***d* of the bottom tray **110**. In FIG. **7**, the carrying handle **140** is partly lowered with the respective movement of the latch **142** upward. As the carrying handle **140** is lowered, the latch **142** moves upward pushing against the top panel **121** to move the top enclosure **120** upward toward the open position. In FIG. **8**, the carrying handle **140** is fully lowered and parallel with the bottom tray **110** with the latch **142** concurrently moving into the upright position, which is perpendicular with the

length of the bottom tray. In this position, the top enclosure **120** is fully open with the latch **142** resting against the left side wall **122***a* and the right side **122***b* and preventing the top enclosure from moving down into the collapsed position. The top enclosure **120** can be readily moved into the collapsed position by lifting the carrying handle **140** which concurrently moves the latch **142**. The top panel **121** can be pressed downward to collapse the top enclosure **120** onto the bottom tray **110**.
(23) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. This description is presented for purposes of illustration and is not intended to be exhaustive or limited to the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the appended claims.

Claims

1. A collapsible animal litter box comprising: a bottom tray having four upstanding walls, an open top, and a closed bottom; a top enclosure comprising a first side wall, a second side wall, and a roof, said roof hingedly connected to the first and second side walls; said first and second walls being hingedly connected to the bottom tray; said first and second walls further comprising hinges that are aligned horizontally in center of the first and second walls; a back wall comprising a top and a bottom, said bottom being hingedly connected to the upstanding walls of the bottom tray; collapsible animal litter box further comprising a handle and a latch; said handle being located outside the four upstanding walls of the bottom tray; said latch being located within the four upstanding walls of the bottom tray; said handle being connected to said latch at a perpendicular angle; said handle and latch being rotatably connected to the bottom tray; whereby when the handle is positioned parallel to the closed bottom of the bottom tray, the latch will push upward on the roof of the top enclosure; whereby when the handle is positioned perpendicular to the closed bottom of the bottom tray, the latch will push recede into the bottom tray, allowing the top enclosure to collapse; whereby the collapsible animal litter box is opened by positioning the handle perpendicular to the closed bottom of the bottom tray and erecting the first and second walls into an upright position and positioning the back wall so that the roof rests on the top of the back wall; whereby the collapsible animal litter box is stored by positioning the handle perpendicular to the closed bottom of the bottom tray and folding the first and second walls inward with the roof resting on top of the folded first and second walls and folding down the back wall.
2. A collapsible animal litter box comprising: a bottom tray having four upstanding walls, an open top, and a closed bottom; a top enclosure comprising a first side wall, a second side wall, and a roof, said roof hingedly connected to the first and second side walls; said first and second walls being hingedly connected to the bottom tray; said first and second walls further comprising hinges that are aligned horizontally in center of the first and second walls; the collapsible animal litter box further comprising a handle and a latch; said handle being located outside the four upstanding walls of the bottom tray; said latch being located within the four upstanding walls of the bottom tray; said handle being connected to said latch at a perpendicular angle; said handle and latch being rotatably connected to the bottom tray; whereby when the handle is positioned parallel to the closed bottom of the bottom tray, the latch will push upward on the roof of the top enclosure; whereby when the handle is positioned perpendicular to the closed bottom of the bottom tray, the latch will push recede into the bottom tray, allowing the top enclosure to collapse; whereby the collapsible animal litter box is opened by using the handle to push upward on the roof of the top enclosure and erecting the first and second walls into an upright position; whereby the collapsible animal litter box is stored by positioning the handle perpendicular to the closed bottom of the bottom tray and

folding the first and second walls inward with the roof resting on top of the folded first and second walls.